

# MOTION TOMOGRAPHY VIA OCCUPATION KERNELS

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**ABSTRACT.** The goal of motion tomography is to recover a description of a vector flow field using information on the trajectory of a sensing unit. In this paper, we develop a predictor corrector algorithm designed to recover vector flow fields from trajectory data with the use of occupation kernels developed by Rosenfeld et al. [9, 10]. Specifically, we use the occupation kernels as an adaptive basis; that is, the trajectories defining our occupation kernels are iteratively updated to improve the estimation on the next stage. Initial estimates are established, then under mild assumptions, such as relatively straight trajectories, convergence is proven using the Contraction Mapping Theorem. We then compare to the established method by Chang et al. [5] by defining a set of error metrics. We found that for simulated data, which provides a ground truth, our method offers a marked improvement and that for a real-world example we have similar results to the established method.

1. **Introduction.** Over the past decade, unmanned aircraft and underwater systems have evolved significantly and are on the verge of becoming a ubiquitous part of urban and littoral landscape. To compensate for the lack of access to the global positioning system (GPS), unmanned underwater vehicles (UUVs) often rely on the knowledge of the flow field to improve localization [8]. Similarly, accurate estimation of urban wind fields is widely acknowledged to be a significant challenge for implementation of traffic control systems (such as [2]) for unmanned air vehicles (UAVs) [11]. While it is possible to model air flow fields and ocean currents using measurements from on-board sensors, lack of accurate localization (in the case of UUVs) and vehicle-induced noise (in the case of UAVs, especially multi-rotor UAVs), creates significant challenges in acquisition and processing of the data generated by on-board sensors. The aforementioned challenges, along with the payload

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reduction associated with removing flow sensors, motivates the development of estimation techniques that rely only on the effect of the flow field on the motion of UAVs and UUVs, and not on direct measurements of the flow velocities.

Motion tomography refers to the reconstruction of a vector field using its accumulated effects on mobile sensing units as they travel through the field [14, 5]. Motion tomography allows for the use of low cost mobile underwater/air vehicles as sensors to accumulate sufficient data for estimation of vector fields resulting from wind and ocean currents. As a result, military applications such as ocean current mapping for effective navigation of mine countermeasure UUVs in littoral environments, commercial applications such as wind field mapping for navigation of small package delivery UAVs, and disaster response applications such as wind field mapping for the prediction of flame front propagation and smoke spread, stand to benefit from fast and accurate motion tomography. In this paper we propose an algorithm for motion tomography based on occupation kernels developed in [9, 10]. We show this algorithm converges via the contraction mapping theorem.

The developed approach to motion tomography has several advantages over existing techniques such as [5]. The flow field is approximated here using the occupation kernels as basis functions for approximation, whereas [5] requires a piecewise constant description of the flow field or a parameterization with respect to Gaussian RBFs. Moreover, [5] employs a renormalization routine which imposes limitations on the motion of the mobile sensors. The proposed occupation kernel method avoids the renormalization and does not add further restrictions on the motion. Finally, the representation of the flow field with respect to the occupation kernel basis allows for the application of the approximation abilities of RKHSs, which are exploited in the convergence analysis.

## 2. Tools.

**Definition 2.1.** A reproducing kernel Hilbert space (RKHS),  $H$ , over a set  $X$  is a Hilbert space of real valued functions over the set  $X$  such that for all  $x \in X$  the evaluation functional  $E_x g := g(x)$  is bounded.

**Remark 1.** Since the evaluation functional is bounded (hence continuous) over a reproducing kernel Hilbert space, the Riesz representation theorem guarantees, for all  $x \in X$ , the existence of a function  $K_x \in H$  such that  $\langle g, K_x \rangle_H = g(x)$ , where  $\langle \cdot, \cdot \rangle_H$  is the inner product for  $H$  [7, Chapter 1]. The function  $K_x$  is called the reproducing kernel function at  $x$ , and  $K_y(x) = K(x, y) = \langle K_y, K_x \rangle_H$ .

**Remark 2.** Unless indicated otherwise all inner products are taken to be the Hilbert space inner product, i.e.  $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_H$ . The same is not true for norms, and differing norms will be indicated by subscripts.

**Remark 3.** Most of this paper is agnostic of the selection of kernel. By Moore–Aronszajn Theorem (cf. [1]) there is a one to one correspondence between symmetric positive semi-definite kernels and reproducing kernel Hilbert spaces. Much of the analysis of the paper and the algorithm will work with any kernel. The two primary examples used in the experiments are the Gaussian Kernels,  $K(x, y) = e^{-\frac{1}{\mu} \|x-y\|_2^2}$ , and exponential dot product kernels,  $K(x, y) = e^{\frac{x \cdot y}{\mu}}$ , in which  $\mu \in \mathbb{R}$  is a tunable hyperparameter often called the “kernel width” in the case of Gaussians. Only for estimates involving the Gram matrices do we specify to Gaussian kernels, where many results from scattered data interpolation are leveraged (cf. [13]).

**Definition 2.2.** Let  $X \subset \mathbb{R}^n$  be compact,  $H$  be a RKHS of continuous functions over  $X$ , and  $\gamma : [0, T] \rightarrow X$  be a bounded measurable trajectory. The functional  $g \mapsto \int_0^T g(\gamma(\tau))d\tau$  is bounded, and may be respresented as  $\int_0^T g(\gamma(\tau))d\tau = \langle g, \Gamma_\gamma \rangle_H$ , for some  $\Gamma_\gamma \in H$  by the Riesz representation theorem. The function  $\Gamma_\gamma$  is called the occupation kernel corresponding to  $\gamma$  in  $H$ .

The value of an inner product against an occupation kernel in a RKHS can be approximated by leveraging quadrature techniques for integration. The numerical experiments described below utilize Simpson's rule when computing inner products involving occupation kernels and their associated Gram matrices, pivotal in the analysis contained in Theorem 4.1. We present Theorem 2.3 which shows convergence with Simpson's Rule. Moreover, the occupation kernels themselves can be expressed as an integral against the kernel function in a RKHS as demonstrated in Proposition 1. Theorem 2.3 and Proposition 1 originally appear in [10], but are reproduced below for completeness.

**Proposition 1.** Let  $H$  be a RKHS over a compact set  $X$  consisting of continuous functions with kernel  $K(x, y)$  and let  $\gamma : [0, T] \rightarrow X$  be a continuous trajectory as in Definition 2.2. The occupation kernel corresponding to  $\gamma$  in  $H$ ,  $\Gamma_\gamma$ , may be expressed as

$$\Gamma_\gamma(x) = \int_0^T K(x, \gamma(t))dt. \quad (1)$$

*Proof.* Note that  $\Gamma_\gamma(x) = \langle \Gamma_\gamma, K(\cdot, x) \rangle_H$ , by the reproducing property of  $K$ . Consequently,

$$\Gamma_\gamma(x) = \langle \Gamma_\gamma, K(\cdot, x) \rangle_H = \langle K(\cdot, x), \Gamma_\gamma \rangle_H = \int_0^T K(\gamma(t), x)dt = \int_0^T K(x, \gamma(t))dt,$$

which establishes the result.  $\square$

Leveraging Proposition 1, quadrature techniques can be demonstrated to give not only pointwise convergence but also norm convergence in the RKHS, which is a strictly stronger result.

**Theorem 2.3.** Under the hypothesis of Proposition 1, let  $t_0 = 0 < t_1 < t_2 < \dots < t_F = T$  (with  $F$  even and  $t_i$  evenly spaced), suppose that  $\gamma$  is a fourth order continuously differentiable trajectory and  $H$  is composed of fourth order continuously differentiable functions. Set  $h$  to satisfy  $t_i = t_0 + ih$ , and consider

$$\hat{\Gamma}_\gamma(x) := \frac{h}{3} \left( K(x, \gamma(t_0)) + 4 \sum_{i=1}^{\frac{F}{2}} K(x, \gamma(t_{2i-1})) + 2 \sum_{i=1}^{\frac{F}{2}-1} K(x, \gamma(t_{2i})) + K(x, \gamma(t_F)) \right). \quad (2)$$

The norm distance is bounded as  $\|\Gamma_\gamma - \hat{\Gamma}_\gamma\|_H^2 = O(h^4)$ .

*Proof.* Consider  $\|\Gamma_\gamma - \hat{\Gamma}_\gamma\|_H^2 = \langle \Gamma_\gamma, \Gamma_\gamma \rangle_H + \langle \hat{\Gamma}_\gamma, \hat{\Gamma}_\gamma \rangle_H - 2\langle \Gamma_\gamma, \hat{\Gamma}_\gamma \rangle_H$ . The term  $\langle \hat{\Gamma}_\gamma, \hat{\Gamma}_\gamma \rangle_H$  is an implementation of the two-dimensional Simpson's rule (cf. [4]) while  $\langle \Gamma_\gamma, \Gamma_\gamma \rangle_H$  is the double integral  $\int_0^T \int_0^T K(\gamma(t), \gamma(\tau))dtd\tau$ . Thus,

$$\langle \hat{\Gamma}_\gamma, \hat{\Gamma}_\gamma \rangle_H = \langle \Gamma_\gamma, \Gamma_\gamma \rangle_H + O(h^4).$$

86 Similarly,  $\langle \Gamma_\gamma, \hat{\Gamma}_\gamma \rangle_H$  integrates in one variable while implementing Simpson's rule  
 87 in the other. Consequently,

$$\langle \Gamma_\gamma, \hat{\Gamma}_\gamma \rangle_H = \langle \Gamma_\gamma, \Gamma_\gamma \rangle_H + O(h^4).$$

88 The conclusion of the theorem follows.  $\square$

89 It should be noted that the above convergence rate is for norm convergence and  
 90 point-wise convergence can be demonstrated to be faster [10].

91 **3. Problem Setup and Developed Algorithm.** Let  $r : [0, T] \rightarrow \mathbb{R}^2$  represent a  
 92 continuous trajectory for a mobile sensor attempting to travel in a straight line, but  
 93 subject to an unknown flow field,  $F : X \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , where  $X$  is a compact subset  
 94 of the plane. Let  $\dot{r} = s \begin{pmatrix} \cos(\theta) & \sin(\theta) \end{pmatrix}^T + F(r)$ , for a positive constant  $s$ , represent  
 95 the true dynamics induced by the flow field. We will assume  $F : X \rightarrow \mathbb{R}^2$  is locally  
 96 Lipschitz in order to assure uniqueness of the solutions [12]. As the flow field is  
 97 unknown, the anticipated dynamics are given as  $\dot{\tilde{r}} = s \begin{pmatrix} \cos(\theta) & \sin(\theta) \end{pmatrix}^T$ . After a  
 98 mobile sensor has traveled through the flow field over a time period  $[0, T]$ , during  
 99 which there is limited to no knowledge of the mobile sensor's position, the difference  
 100 between the actual location of the mobile sensor,  $r(T)$ , and the anticipated location,  
 101  $\tilde{r}(T)$ , is given as

$$D = r(T) - \tilde{r}(T) = \int_0^T (\dot{r}(t) - \dot{\tilde{r}}(t)) dt = \int_0^T F(r(t)) dt$$

102 Hence,  $D = \langle F, \Gamma_r \rangle_H$  and the difference between the final locations of the mobile  
 103 sensor describes the integral of the flow field along the trajectory  $r$ . This integral  
 104 provides a type of tomographic sample of  $F$ . As the trajectory  $r$  is treated as  
 105 unknown, an approximation of  $F$  using the sample generated by  $\Gamma_r$  is difficult to  
 106 assess directly. This motivates the developed iterative algorithm to determine the  
 107 flow field,  $F$ , as well as the true trajectories,  $r$ . **On each iteration, we use the**  
 108 **current approximation to calculate a simulated trajectory. The difference between**  
 109 **the end points of the simulated and the real trajectory is then used to update our**  
 110 **vector-field through solving a matrix equation. Using notation to be established**  
 111 **below: For each  $i = 1, \dots, s$ ,**

$$w_{1,n} \int_0^T \Gamma_{\tilde{r}_{1,n}}(\tilde{r}_{1,n}(t)) dt + \dots + w_{s,n} \int_0^T \Gamma_{\tilde{r}_{s,n}}(\tilde{r}_{s,n}(t)) dt = D_{i,n} + \int_0^T \hat{F}_n(\tilde{r}_{i,n}(t)) dt.$$

112 In this form, we see that the  $D_{i,n}$  acts as a correction term.

113 Let  $\{s_i\}_{i=1}^M, \{\theta_i\}_{i=1}^M$  be a collection of speeds and angles used to generate a  
 114 collection of anticipated trajectories, i.e. trajectories governed by the dynamics  
 115  $\dot{\tilde{r}} = s_i \begin{pmatrix} \cos(\theta_i) & \sin(\theta_i) \end{pmatrix}^T$ . Moreover, let  $\{p_i\}_{i=1}^M \subset \mathbb{R}^2$  represent the starting point  
 116 of the trajectories.

117 Significantly, after the initial data collection period, no further experiments are  
 118 necessary to approximate the flow field. Specifically, Algorithm 1 only needs to  
 119 produce new simulations of the approximate trajectories which ultimately converge  
 120 to the true trajectories from the initial experiment.

**Algorithm 1** Iterative Motion Tomography Algorithm

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Define  $N$  as the number of iterates  
Input: Samples  $r_i(T)$   $i \in \{1, \dots, s\}$   
Set  $D_{i,0} = r_i(T) - \tilde{r}_{i,0}(T)$   $i \in \{1, \dots, s\}$   
Set  $\hat{F}_0 = 0$   
**for**  $n \in \{0, \dots, N\}$  **do**  
  **for**  $i \in \{1, \dots, s\}$  **do**  
    Generate, via a numerical method,  $\tilde{r}_{i,n} : [0, 1] \rightarrow X$  as the unique solution to  

$$\dot{p} = s_i \begin{pmatrix} \cos(\theta_i) & \sin(\theta_i) \end{pmatrix}^\top + \hat{F}_n(p), \quad p(0) = p_i.$$
  
  **end for**  
  Set  $D_{i,n} = r_i(T) - \tilde{r}_{i,n}(T)$   $i \in \{1, \dots, s\}$   
  **for**  $i \in \{1, \dots, s\}$  **do**  
    Compute  $\{w_{i,n+1}\}_{i=1}^s$  by solving  

$$\left( \langle \Gamma_{\tilde{r}_{i,n}}, \Gamma_{\tilde{r}_{j,n}} \rangle \right)_{i,j=1}^{s,s} \begin{pmatrix} w_{1,n+1} \\ \vdots \\ w_{s,n+1} \end{pmatrix} = \begin{pmatrix} D_{1,n} + \langle \hat{F}_n, \Gamma_{\tilde{r}_{1,n}} \rangle \\ \vdots \\ D_{M,n} + \langle \hat{F}_n, \Gamma_{\tilde{r}_{s,n}} \rangle \end{pmatrix}.$$
  
  **end for**  
  Output  $\hat{F}_{n+1} = \sum_{i=1}^s w_{i,n+1} \Gamma_{\tilde{r}_{i,n}}$   
**end for**

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121 **4. Convergence of Algorithm.** This section is devoted to establishing sufficient  
122 conditions for the above algorithm to converge. The main theoretical crux will be  
123 the contraction mapping theorem. Let  $X \subset \mathbb{R}^2$  be compact,  $F : X \rightarrow \mathbb{R}^2$  a Lipschitz  
124 continuous vector field, let  $\{r_i\}_{i=1}^\infty$  be a countable set of trajectories,  $r_i : [0, T] \rightarrow X$ ,  
125 satisfying

$$\dot{p} = s_i \begin{pmatrix} \cos(\theta_i) & \sin(\theta_i) \end{pmatrix}^\top + F(p), \quad p(0) = p_i$$

126 for some countable collection of  $(s_i, \theta_i) \in \mathbb{R} \times [0, 2\pi)$ , and countable set  $\{p_i\}_{i=1}^\infty \subset X$ .  
127 Let  $H(X)$  be a reproducing kernel Hilbert space of  $\mathbb{R}^2$  valued functions.

128 **Theorem 4.1.** Let  $\Omega \subset H(X)$  be the subset of  $\phi(x) = (\phi_1(x) \ \phi_2(x))^\top$  with  
129  $\phi_i(x) = \sum_{j=1}^s w_{j,\phi}^i \Gamma_{r_{j,\phi}}(x)$  for  $i = 1, 2$ . We define  $A : \Omega \subset H(X) \rightarrow H(X)$  as  
130 follows: Given,  $\phi \in \Omega$  define  $\hat{r}_{j,\phi}$  to be the solution to

$$\dot{p} = s_j \begin{pmatrix} \cos(\theta_j) & \sin(\theta_j) \end{pmatrix}^\top + \phi(p) \quad p(0) = p_j.$$

131 Moreover, define  $D_{j,\phi}^i := r_j^i(T) - \hat{r}_{j,\phi}^i(T)$  for  $i = 1, 2$ . Then

$$A(\phi)_i := \sum_{j=1}^s \hat{w}_{j,\phi}^i \Gamma_{\hat{r}_{j,\phi}}(x)$$

132 where  $\hat{w}_{j,\phi}^i$  are defined so that  $A(\phi)_i$  satisfies

$$\langle A(\phi)_i, \Gamma_{\hat{r}_{j,\phi}} \rangle_H = D_{j,\phi}^i + \langle \phi, \Gamma_{\hat{r}_{j,\phi}} \rangle_H$$

133 for  $i = 1, 2$ . Under the set of assumptions, (1) – (2), listed in following section  
134 there exists a closed finite diameter subspace  $E \subset \Omega$  such that  $A|_E$  is a contraction  
135 mapping and thus extends to a contraction mapping on  $H(X)$ .

We note that Hilbert spaces have the Lipschitz extension property [3], i.e. a Banach space  $E$  has the Lipschitz (Contraction) extension property if every Lipschitz map on a subset of  $E$  extended to all of  $E$  with the same Lipschitz constant. Moreover, we will also not be proving existence of the solution, but rather We will be proving that the mapping described above is a contraction in some ball of diameter less than one containing the solution and applying a version of Contraction Mapping Theorem (cf. [6]) to prove convergence. Furthermore, we will also be suppressing the index  $i = 1, 2$  and the use of operator valued kernels to facilitated the simplicity of presentation, and it should be noted that the proof extends to any finite size dimension. Note, the operator valued kernels being used are of the form  $\mathbb{K}(x, y) = \text{diag}(K(x, y))$ .

**Theorem 4.2** (Contraction Mapping Theorem). *Let  $X$  be a complete metric space and let  $S_0$  be a closed subset of  $X$  of finite diameter. Let  $P : S_0 \rightarrow S_0$  be a contraction mapping. Then the sequence of iterates  $\{x_k\}$  produced by successive iterations  $x_{k+1} = P(x_k)$  converges to  $x = P(x)$ , the unique fixed point of  $P$  in  $S_0$  for any  $x_0$  in  $S_0$ .*

**Remark 4.** The contraction mapping theorem guarantees convergence in the metric. As the metric space is a reproducing kernel Hilbert space the metric is given as a norm difference. Hence, the convergence is in terms of the norm, which for reproducing kernel Hilbert spaces gives point-wise convergence.

To prove Theorem 4.1 we establish several propositions that give inequalities relating norms to the norms of the flow fields.

**Proposition 2.** *Let  $H(X)$  be the reproducing kernel Hilbert space described above. For  $\phi, \psi \in H(X)$ , if  $r_\phi$  and  $r_\psi$  are the unique solutions to the initial value problems*

$$\begin{aligned} \dot{p} &= s \begin{pmatrix} \cos(\theta) & \sin(\theta) \end{pmatrix}^\top + \phi(p), \quad p(0) = p \\ \dot{p} &= s \begin{pmatrix} \cos(\theta) & \sin(\theta) \end{pmatrix}^\top + \psi(p), \quad p(0) = p \in X \end{aligned}$$

respectively, then

$$|r_\phi(t) - r_\psi(t)| \leq M \|\phi - \psi\|_H$$

for some constant  $M$ .

*Proof.* We have that

$$\begin{aligned} |\dot{r}_\phi - \dot{r}_\psi| &= |\phi(r_\phi(t)) - \psi(r_\psi(t))| \leq |\phi(r_\phi(t)) - \phi(r_\psi(t))| + |\phi(r_\psi(t)) - \psi(r_\psi(t))| \\ &\leq \|\nabla \phi\|_\infty \cdot |r_\phi(t) - r_\psi(t)| + \|\phi - \psi\|_H \sqrt{K(r_\psi(t), r_\psi(t))} \end{aligned}$$

by an application of mean value theorem. Note that  $\|\cdot\|_\infty$  denotes the sup-norm.

It follows that,

$$|r_\phi(t) - r_\psi(t)| \leq \exp(\|\nabla \phi\|_\infty) |r_\phi(0) - r_\psi(0)| + \|\phi - \psi\|_H \cdot M = \|\phi - \psi\|_H \cdot M$$

for  $M = \max_x K(x, x)$  which exists by compactness of the region.  $\square$

The proof of the following proposition can be found in [10], but is presented here for clarity.

**Proposition 3.** *Suppose  $H$  is a RKHS over a set  $X$  consisting of continuous functions and let  $\gamma_1(t)$  and  $\gamma_2(t)$  be two trajectories with homotopy  $\{\gamma_w(t)\}$ . The map  $w \mapsto \Gamma_{\gamma_w}$  is continuous.*

171 *Proof.* As  $[0, T] \times [0, 1]$  is compact, the map  $(t, w) \mapsto \gamma_w(t)$  is uniformly continu-  
 172 ous. That is for every  $\varepsilon > 0$  there exists a  $\delta > 0$  such that whenever  $\|(t_1, w_1) -$   
 173  $(t_2, w_2)\|_2 < \delta$ ,  $\|\gamma_{w_1}(t_1) - \gamma_{w_2}(t_2)\|_2 < \varepsilon$ . Moreover, as  $K(\cdot, \cdot)$  is continuous, and the  
 174 image of  $\gamma_w(t)$  is compact, the map  $(w_1, t, w_2, \tau) \mapsto K(\gamma_{w_1}(t), \gamma_{w_2}(\tau))$  is uniformly  
 175 continuous.

Fix  $\varepsilon > 0$  and select  $\delta > 0$  such that

$$\begin{aligned} |K(\gamma_{w_1}(t), \gamma_{w_1}(\tau)) - K(\gamma_{w_2}(t), \gamma_{w_1}(\tau))| &< \frac{\varepsilon}{2T^2} \quad \text{and} \\ |K(\gamma_{w_2}(t), \gamma_{w_2}(\tau)) - K(\gamma_{w_2}(t), \gamma_{w_1}(\tau))| &< \frac{\varepsilon}{2T^2} \end{aligned}$$

whenever  $|w_1 - w_2| < \delta$ . Select  $w_1, w_2$  such that  $|w_1 - w_2| < \delta$ , then

$$\begin{aligned} \|\Gamma_{w_1} - \Gamma_{w_2}\|_H^2 &= \langle \Gamma_{w_1}, \Gamma_{w_1} \rangle + \langle \Gamma_{w_2}, \Gamma_{w_2} \rangle - 2\langle \Gamma_{w_1}, \Gamma_{w_2} \rangle \\ &= \int_0^T \int_0^T (K(\gamma_{w_1}(t), \gamma_{w_1}(\tau)) - K(\gamma_{w_2}(t), \gamma_{w_1}(\tau))) dt d\tau \\ &\quad + \int_0^T \int_0^T (K(\gamma_{w_2}(t), \gamma_{w_2}(\tau)) - K(\gamma_{w_2}(t), \gamma_{w_1}(\tau))) dt d\tau, \end{aligned} \quad (3)$$

176 Note that Equation (3) is positive and bounded by  $\varepsilon$  by construction. Hence, the  
 177 map  $w \mapsto \Gamma_w$  is continuous.  $\square$

178 **Corollary 1.** *Following the notational convention defined in Proposition 3, if*  
 179  *$|r_\phi(t) - r_\psi(t)| \leq M\|\phi - \psi\|_H$  for some  $M$  then  $\|\Gamma_{r_\phi} - \Gamma_{r_\psi}\|_H \leq \sqrt{CT}\|\phi - \psi\|_H^{1/2}$*   
 180 *as well for some constant  $C$  independent of  $T$ .*

*Proof.*

$$\begin{aligned} \|\Gamma_{r_\phi} - \Gamma_{r_\psi}\|_H^2 &= \langle \Gamma_{r_\phi}, \Gamma_{r_\phi} \rangle + \langle \Gamma_{r_\psi}, \Gamma_{r_\psi} \rangle - 2\langle \Gamma_{r_\phi}, \Gamma_{r_\psi} \rangle \\ &= \int_0^T \int_0^T K(r_\phi(t), r_\phi(\tau)) - K(r_\psi(t), r_\phi(\tau)) dt d\tau \\ &\quad + \int_0^T \int_0^T K(r_\psi(t), r_\psi(\tau)) - K(r_\psi(t), r_\phi(\tau)) dt d\tau \\ &\leq \int_0^T \int_0^T (|\nabla_y K(r_\phi(t), \xi_\psi)|) \cdot |r_\phi(\tau) - r_\psi(\tau)| dt d\tau \\ &\quad + \int_0^T \int_0^T (|\nabla_y K(r_\psi(t), \xi_\phi)|) \cdot |r_\phi(\tau) - r_\psi(\tau)| dt d\tau \\ &\leq CT^2 \|\phi - \psi\|_H. \end{aligned} \quad (4)$$

181 The last two inequalities are due to an application of mean value theorem and  
 182 proposition 2.  $\square$

183 In order to talk about the stability of the interpolation process we define the  
 184 following:

185 **Definition 4.3.** The trajectory separation distance is given by

$$q_{X,T} := \frac{1}{2} \min_{\substack{j \neq k \\ t, \tau \in [0, T]}} \|\gamma_j(t) - \gamma_k(\tau)\|_2$$

186 This is the maximum radius such that all “tubes” centered on the trajectories are  
 187 disjoint.

188 The next theorem will be a trajectory variant of a theorem found in Wendland  
 189 [13]. In Wendland, a bound for the minimum eigenvalue is obtained by expressing  
 190 the kernel function  $\Phi(x - y)$  in terms of its Fourier transform and comparing with  
 191 an intermediate radial functions  $\Psi_M$  based on characteristic functions for balls of  
 192 radius  $M$ . This comparison is given in terms of the separation distance between  
 193 centers. By defining the trajectory separation distance as above the inequalities  
 194 found in Wendland remain valid since the points in the image of each trajectory are  
 195 separated by at least  $q_{X,T}$ .

196 In the following theorem let  $G_\Gamma$  be the Occupation kernel Grammian, i.e. for a  
 197 set of trajectories,  $r_i : [0, T] \rightarrow \mathbb{R}^2$ , with  $1 \leq i \leq N$

$$G_\Gamma := (\langle \Gamma_{r_j}, \Gamma_{r_k} \rangle)_{j,k=1}^N = \int_0^T \int_0^T (K(r_j(t), r_k(\tau)))_{j,k=1}^N dt d\tau.$$

198 We also need the following definition found in [13]:

199 **Definition 4.4.** We say a function  $f$  is slowly increasing if there exists a constant  
 200  $m \in \mathbb{N}_0$  such that  $f(x) = \mathcal{O}(\|x\|_2^m)$  for  $\|x\|_2 \rightarrow \infty$ .

201 **Remark 5.** Essentially a function is “slowly increasing” if it is polynomially bounded  
 202 for large  $x$ . As Gaussians decay to zero as  $\|x\|_2 \rightarrow \infty$ , Gaussians are bounded by a  
 203 constant and fall under this category.

204 **Theorem 4.5.** With  $\Phi(\|x - y\|_2) = K(x, y) = \exp(-\mu\|x - y\|_2^2)$  the minimal ei-  
 205 genvalue of the Occupation kernel Grammian is bounded by

$$\lambda_{\min}(G_\Gamma) \geq \frac{C_2 \exp(-M_2^2/(q_{X,T}^2 \mu)) T^2}{2\mu q_{X,T}^2}$$

206 with  $M_2 = 12 \sqrt[3]{\frac{\pi}{9}}$  and  $C_2 = \frac{M^2}{16}$ .

207 *Proof.* Eventually, we will have  $d = 2$  but for now we will write generally. Let  $\Phi :$   
 208  $\mathbb{R}^d \rightarrow \mathbb{C}$  be any radial, continuous, and slowly increasing function possessing a  
 209 non-negative generalized Fourier transform linearity, for  $\alpha = (\alpha_1, \dots, \alpha_s)^\top$ , given,

$$\alpha^\top G_{\Phi(r)} \alpha = \sum_{j,k=1}^s \alpha_j \alpha_k \Phi(r_j(t) - r_k(\tau)) = \int_{\mathbb{R}^d} \sum_{j,k=1}^s \alpha_j \alpha_k e^{i\omega^\top (r_j(t) - r_k(\tau))} \hat{\Phi}(\omega) d\omega$$

by linearity an application of Fubini we have,

$$\begin{aligned} \alpha^\top G_\Gamma \alpha &= \int_0^T \int_0^T \alpha^\top G_{\Phi(r)} \alpha dt d\tau \\ &= \int_0^T \int_0^T \int_{\mathbb{R}^d} \sum_{j,k=1}^s \alpha_j \alpha_k e^{i\omega^\top (r_j(t) - r_k(\tau))} \hat{\Phi}(\omega) d\omega \\ &= \int_{\mathbb{R}^d} \left| \sum_{j=1}^s \alpha_j \int_0^T e^{i\omega^\top r_j(t)} dt \right|^2 \hat{\Phi}(\omega) d\omega \end{aligned} \tag{5}$$

210 where (5) is valid whenever  $\Phi : \mathbb{R}^d \rightarrow \mathbb{C}$  is a continuous and slowly increasing  
 211 function possessing a non-negative generalized Fourier transform. Define,

$$\psi_{M_2}(x) = \frac{\varphi_0(M_2) \Gamma\left(\frac{d}{2} + 1\right)}{2^{d/2}} \|x\|_2^{-d} J_{d/2}^2(M_2 \|x\|_2)$$



212 where  $J_\nu$  is the Bessel function of the first kind and

$$\varphi_0(M_2) = \inf_{\|\omega\|_2 \leq 2M_2} \hat{\Phi}(\omega).$$

In Wendland [13] it is established that for a collection of points  $x_i, i = 1, \dots, s$ ,

$$\begin{aligned} \int_{\mathbb{R}^d} \sum_{j,k=1}^s \alpha_j \alpha_k e^{i\omega^\top (x_j - x_k)} \hat{\psi}_{M_2}(\omega) d\omega &= \sum_{j,k=1}^s \alpha_j \alpha_k \psi_{M_2}(x_j - x_k) \\ &\geq \|\alpha\|_2^2 \left( \psi_{M_2}(0) - \max_{1 \leq j \leq s} \sum_{k=1, k \neq j}^s |\psi_{M_2}(x_j - x_k)| \right) \end{aligned}$$

213 and that for the given value of  $M_2$ ,

$$\max_{1 \leq j \leq s} \sum_{k=1, k \neq j}^s |\psi_{M_2}(x_j - x_k)| \leq \frac{1}{2} \psi_{M_2}(0).$$

214 Thus, for the stated value of  $M_2$  we have that,

$$\sum_{j,k=1}^s \alpha_j \alpha_k \psi_{M_2}(x_j - x_k) \geq \|\alpha\|_2^2 \frac{\psi_{M_2}(0)}{2}. \quad (6)$$

215 Now, the key insight is that the establishment of 6 is unaffected by letting  $x_j =$   
216  $r_j(t), x_k = r_k(\tau)$  for  $t, \tau \in [0, T]$  by replacing the standard separation distance  $q_X$   
217 with  $q_{X,T}$  and that this holds over  $[0, T]$ .

218 Our auxiliary function  $\psi_{M_2}$  is designed so that  $\hat{\psi}_{M_2}(\omega) \leq \hat{\Phi}(\omega)$ , giving us

$$\int_{\mathbb{R}^d} \left| \sum_{j=1}^s \alpha_j \int_0^T e^{i\omega^\top r_j(t)} dt \right|^2 \hat{\Phi}(\omega) d\omega \geq \int_{\mathbb{R}^d} \left| \sum_{j=1}^s \alpha_j \int_0^T e^{i\omega^\top r_j(t)} dt \right|^2 \hat{\psi}_{M_2}(\omega) d\omega. \quad (7)$$

By combining equations, (5), (5), (6), and (7) we get

$$\begin{aligned} \alpha^\top G_\Gamma \alpha &= \int_0^T \int_0^T \alpha^\top G_{\Phi(r)} \alpha dt d\tau = \int_0^T \int_0^T \int_{\mathbb{R}^d} \sum_{j,k=1}^s \alpha_j \alpha_k e^{i\omega^\top (r_j(t) - r_k(\tau))} \hat{\Phi}(\omega) d\omega \\ &\geq \int_0^T \int_0^T \|\alpha\|_2^2 \frac{\psi_{M_2}(0)}{2} dt d\tau. \end{aligned}$$

219 Finally,

$$\lambda_{\min}(G_\Gamma) = \inf \frac{\alpha^\top G_\Gamma \alpha}{\|\alpha\|_2^2} \geq \frac{C_2 \exp(-M_2^2/(q_{X,T}^2 \mu)) T^2}{2\mu q_{X,T}^2}$$

220 with  $M_2 = 12 \sqrt[3]{\frac{\pi}{9}}$  and  $C_2 = \frac{M_2^2}{16}$  and using the bounds for  $\psi_M(0)$  established in  
221 [13].  $\square$

222 As a corollary to the above analysis we show that the condition number of the  
223 interpolation matrices is at least as good as the standard Gaussian kernels.

224 **Corollary 2.** *If*

$$G_\Gamma = (\langle \Gamma_{r_j}, \Gamma_{r_k} \rangle)_{j,k=1}^{s,s} = \int_0^T \int_0^T (K(r_j(t), r_k(\tau)))_{j,k=1}^{s,s} dt d\tau,$$

225 *then*

$$\lambda_{\max}(G_\Gamma) \leq sT^2 \Phi(0)$$

where  $\Phi(\|x_i - x_j\|_2) = K(x_j, x_i)$ .

*Proof.* By a standard application of Gershgorin's circle theorem on eigenvalues,

$$\lambda_{\max}(G_\Gamma) \leq s \max_{i,j=1,\dots,s} |\Phi(r_j(t) - r_k(\tau))| \leq s \int_0^T \int_0^T |\Phi(0)| dt d\tau = sT^2 |\Phi(0)|$$

where the above is due to the properties of positive semi-definite functions.  $\square$

Since the condition number of an interpolation matrix  $A$  is given by the ratio of the maximum and minimum eigenvalues, the presence of the  $T^2$  in the above inequality nullifies the  $1/T^2$  appearing in estimate for the minimal eigenvalue. In essence, this shows that can take the minimal distance between trajectories as a measurement of how good our approximations will be.

**4.1. Main Inequalities.** Following the notion established in 4.1 consider the following,

$$\begin{aligned} \|A(\phi) - A(\psi)\|_H &= \left\| \sum_{j=1}^s \hat{w}_{j,\phi} \Gamma_{\hat{r}_{j,\phi}}(x) - \sum_{j=1}^s \hat{w}_{j,\psi} \Gamma_{\hat{r}_{j,\psi}}(x) \right\|_H \\ &\leq \sum_{j=1}^s |\hat{w}_{j,\phi}| \|\Gamma_{\hat{r}_{j,\phi}} - \Gamma_{\hat{r}_{j,\psi}}\|_H + \sum_{j=1}^s |\hat{w}_{j,\phi} - \hat{w}_{j,\psi}| \|\Gamma_{\hat{r}_{j,\psi}}\|_H \end{aligned} \quad (8)$$

To proceed forward, we will need to establish control on terms above in as a function of  $\|\phi - \psi\|_H$ . Note by definition  $(\hat{w}_{1,\phi} \cdots \hat{w}_{s,\phi})^\top$  satisfies

$$\begin{pmatrix} \langle \Gamma_{\hat{r}_{1,\phi}}, \Gamma_{\hat{r}_{1,\phi}} \rangle & \cdots & \langle \Gamma_{\hat{r}_{s,\phi}}, \Gamma_{\hat{r}_{1,\phi}} \rangle \\ \vdots & \ddots & \vdots \\ \langle \Gamma_{\hat{r}_{1,\phi}}, \Gamma_{\hat{r}_{M,\phi}} \rangle & \cdots & \langle \Gamma_{\hat{r}_{M,\phi}}, \Gamma_{\hat{r}_{M,\phi}} \rangle \end{pmatrix} \begin{pmatrix} \hat{w}_{1,\phi} \\ \vdots \\ \hat{w}_{s,\phi} \end{pmatrix} = \begin{pmatrix} D_{1,\phi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle \\ \vdots \\ D_{s,\phi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle \end{pmatrix}. \quad (9)$$

For notational convenience, given a  $\phi \in H(X)$  let

$$G_\phi = \begin{pmatrix} \langle \Gamma_{\hat{r}_{1,\phi}}, \Gamma_{\hat{r}_{1,\phi}} \rangle & \cdots & \langle \Gamma_{\hat{r}_{s,\phi}}, \Gamma_{\hat{r}_{1,\phi}} \rangle \\ \vdots & \ddots & \vdots \\ \langle \Gamma_{\hat{r}_{1,\phi}}, \Gamma_{\hat{r}_{s,\phi}} \rangle & \cdots & \langle \Gamma_{\hat{r}_{s,\phi}}, \Gamma_{\hat{r}_{s,\phi}} \rangle \end{pmatrix}$$

Now,

$$\left\| \begin{pmatrix} \hat{w}_{1,\phi} \\ \vdots \\ \hat{w}_{s,\phi} \end{pmatrix} - \begin{pmatrix} \hat{w}_{1,\psi} \\ \vdots \\ \hat{w}_{s,\psi} \end{pmatrix} \right\|_2 \leq \|G_\phi^{-1}\|_{\text{op}} \left\| \begin{pmatrix} D_{1,\phi} - D_{1,\psi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\phi} - D_{s,\psi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2 \quad (10)$$

$$+ \|G_\phi^{-1} - G_\psi^{-1}\|_{\text{op}} \left\| \begin{pmatrix} D_{1,\psi} + \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\psi} + \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2, \quad (11)$$

where  $\|\cdot\|_{\text{op}}$  denotes the operator norm induced by the 2-norm. A portion of the proof for Theorem 4.1 will be on establishing suitable control over of the norms in (10), and (11). Before proceeding, we will need the following lemma.

240 **Lemma 4.6.** *Given a  $\phi, \psi \in H(X)$ , then*

$$\|G_\phi - G_\psi\|_{\text{op}} \leq ST^2 \|\phi - \psi\|_H^{1/2}$$

241 *for a constant  $S \propto s$  and independent of  $T$*

*Proof.* Similar to the analysis in corollary 2

$$\begin{aligned} \lambda_{\max}(G_\phi - G_\psi) &\leq s \max_{i,j=1,\dots,s} |\langle \Gamma_{\hat{r}_{i,\phi}} - \Gamma_{\hat{r}_{i,\psi}}, \Gamma_{\hat{r}_{j,\phi}} - \Gamma_{\hat{r}_{j,\psi}} \rangle| \\ &\leq s \max_{i,j=1,\dots,s} \|\Gamma_{\hat{r}_{i,\phi}} - \Gamma_{\hat{r}_{i,\psi}}\|_H \cdot \|\Gamma_{\hat{r}_{j,\phi}} - \Gamma_{\hat{r}_{j,\psi}}\|_H \\ &\leq sCT^2 \|\phi - \psi\|_H. \end{aligned}$$

242 Hence, by an application of corollary 1 we have

$$\lambda_{\max}(G_\phi - G_\psi) \leq sCT^2 \|\phi - \psi\|_H^{1/2}.$$

243 Letting  $S = sC$  completes our proof.  $\square$

244 **Analysis of the upper bounds:** Given that by definition

$$|D_{j,\phi} - D_{j,\psi}| = |\hat{r}_{j,\phi}(T) - \hat{r}_{j,\psi}(T)| \leq M \cdot \|\phi - \psi\|_H$$

245 by Proposition 2 and

$$|\langle \phi, \Gamma_{\hat{r}_{j,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{j,\psi}} \rangle| \leq \|\phi\|_H \|\Gamma_{\hat{r}_{j,\phi}} - \Gamma_{\hat{r}_{j,\psi}}\|_H + \|\phi - \psi\|_H \cdot \|\Gamma_{\hat{r}_{j,\psi}}\|_H$$

246 Note that,

$$\left\| \begin{pmatrix} D_{1,\phi} - D_{1,\psi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\phi} - D_{s,\psi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2^2 \leq \sum_{j=1}^s |D_{j,\phi} - D_{j,\psi}|^2 + |\langle \phi, \Gamma_{\hat{r}_{j,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{j,\psi}} \rangle|^2.$$

247 Hence,

$$\left\| \begin{pmatrix} D_{1,\phi} - D_{1,\psi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\phi} - D_{s,\psi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2 \leq \sum_{j=1}^s |D_{j,\phi} - D_{j,\psi}| + |\langle \phi, \Gamma_{\hat{r}_{j,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{j,\psi}} \rangle|$$

since for positive quantities  $x_i$ ,  $\sqrt{\sum_i x_i^2} \leq \sum_i |x_i|$ . From inequality 10

$$\begin{aligned} &\|G_\phi^{-1}\|_{\text{op}} \left\| \begin{pmatrix} D_{1,\phi} - D_{1,\psi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\phi} - D_{s,\psi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle - \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2 \\ &\leq \frac{2\mu}{C_2} \frac{q_{X,T,\phi}^2}{\exp(-M_2^2/(q_{X,T,\phi}^2\mu))T^2} \\ &s \left[ M\|\phi - \psi\|_H + \|\phi\|_H \sqrt{CT} \|\phi - \psi\|_H^{1/2} + \max \|\Gamma_{j,\psi}\|_H \|\phi - \psi\|_H \right] \quad (12) \end{aligned}$$

248 By an application of Corollary 1 and Theorem 4.5. Here we have used  $q_{X,T,\phi}$  to  
249 denote the separation distance for the  $\phi$ -related trajectories.

Turning our attention to inequality (11),

$$\begin{aligned}
& \|G_\phi^{-1} - G_\psi^{-1}\|_{\text{op}} \left\| \begin{pmatrix} D_{1,\psi} + \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\psi} + \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2 \\
& \leq \|G_\phi\|_{\text{op}}^{-1} \|G_\phi - G_\psi\|_{\text{op}} \|G_\psi\|_{\text{op}}^{-1} \cdot \left\| \begin{pmatrix} D_{1,\psi} + \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\psi} + \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2 \\
& \leq \frac{4\mu^2}{C_2^2} \frac{q_{X,T,\phi}^2 q_{X,T,\psi}^2}{\exp(-(M_2^2/q_{X,T,\phi}^2\mu) + (M_2^2/q_{X,T,\psi}^2\mu))T^4} \cdot \\
& sCT^2 \|\phi - \psi\|_H \left\| \begin{pmatrix} D_{1,\psi} + \langle \psi, \Gamma_{\hat{r}_{1,\psi}} \rangle \\ \vdots \\ D_{s,\psi} + \langle \psi, \Gamma_{\hat{r}_{s,\psi}} \rangle \end{pmatrix} \right\|_2
\end{aligned} \tag{13}$$

**Sufficient conditions for convergence:** In order to prove convergence of our algorithm we will make some assumptions. Let  $1 > \varepsilon > 0$ , we will first assume that our iterative procedure starts reasonably close to the true solution  $F$ . In terms of the above, we are going to make the assumption that for both  $\phi$  and  $\psi$  we have  $\|\phi - F\|_H \leq \varepsilon/2$ ,  $\|\psi - F\|_H \leq \varepsilon/2$  and  $\|\phi - \psi\|_H \leq \varepsilon < 1$ . This in turn gives us that  $|D_{j,\phi}|, |D_{j,\psi}| < M\varepsilon$ . We will also make an assumption on our underlying vector field. We are going to assume that the vector field  $F$  has at-least the smoothness conditions stated in section 3 and is not overly powerful in comparison to the mobile sensor engines. In terms of the above, given the differential equation

$$\dot{p} = s_j \begin{pmatrix} \cos(\theta_j) & \sin(\theta_j) \end{pmatrix}^\top + F(p) \quad p(0) = p_j$$

with solution  $r_F(t)$ , we will assume  $\|\Gamma_{r_F} - \Gamma_\Upsilon\|_H < \varepsilon$  where  $\Upsilon(t)$  is the straight line solution to the differential equation above **without the  $F$ -term**, and that  $\|\Gamma_\Upsilon\|$  is bounded by some constant independent of  $T$ . The latter assumption is not unreasonable, as  $\|\Gamma_\Upsilon\|$  would be bounded by the norm of an occupation kernel for the straight line trajectory spanning the diameter of our feature space. With these assumptions, we have  $\|\Gamma_{j,\phi}\|_H, \|\Gamma_{j,\psi}\|_H \leq \|\Gamma_\Upsilon\|_H + \varepsilon$ . Moreover, by Cauchy Schwarz on  $|\langle \phi, \Gamma_{\hat{r}_{j,\phi}} \rangle|$  we have  $\left\| \begin{pmatrix} D_{1,\phi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle, \dots, D_{s,\psi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle \end{pmatrix}^\top \right\|_2$  is bounded by some constant  $L$ . This above discussion culminates in the following assumptions and lemma.

**Assumptions 1.** Let  $F$  be the true vector field. We make the following assumptions:

- 1) The functions  $\phi$  and  $\psi$  as they appear in the above sections satisfy  $\|\phi - F\|_H \leq \varepsilon/2$  and  $\|\psi - F\|_H \leq \varepsilon/2$  for some  $1 > \varepsilon > 0$
- 2) If  $r_F(t)$  is a trajectory given by the true vectorfield  $F$ , then  $\|\Gamma_{r_F} - \Gamma_\Upsilon\| < \varepsilon$  where  $\Gamma_\Upsilon$  is the occupation kernel for the straight line trajectory  $\Upsilon(t)$  starting at the same point as  $r_F(t)$ .
- 3) For all straight line trajectories,  $\Upsilon(t)$ , the norm  $\|\Gamma_\Upsilon\|$  is bounded by some constant independent of  $T$ .

277 **Lemma 4.7.** *With the above assumptions,*

$$\left\| \begin{pmatrix} \hat{w}_{1,\phi} \\ \vdots \\ \hat{w}_{s,\phi} \end{pmatrix} \right\|_2 \leq \|G_\phi^{-1}\|_{op} \left\| \begin{pmatrix} D_{1,\phi} + \langle \phi, \Gamma_{\hat{r}_{1,\phi}} \rangle \\ \vdots \\ D_{s,\phi} + \langle \phi, \Gamma_{\hat{r}_{s,\phi}} \rangle \end{pmatrix} \right\|_2 \leq L \cdot \frac{2\mu}{C_2} \frac{q_{X,T,\phi}^2}{\exp(-M_2^2/(q_{X,T,\phi}^2\mu))T^2}$$

278 *where  $L$  is independent of  $T$ .*

279 **Assumptions 2.** Our final assumption is that

$$\|\Gamma_{r_\phi} - \Gamma_{r_\psi}\|_H \leq CT\|\phi - \psi\|_H$$

280 *for some constant  $C$  independent of  $T$ .*

281 This is an assumption on the regularity on the solutions to the differential equa-  
 282 tions stated in terms of the occupation kernels. While more work is needed, its  
 283 possible that this condition could be stated in terms of Frechet differentiability.

284 **Proof of Theorem 4.1:**

*Proof.* Consider,

$$\begin{aligned} \|A(\phi) - A(\psi)\|_H &= \left\| \sum_{j=1}^s \hat{w}_{j,\phi} \Gamma_{\hat{r}_{j,\phi}}(x) - \sum_{j=1}^s \hat{w}_{j,\psi} \Gamma_{\hat{r}_{j,\psi}}(x) \right\|_H \\ &\leq \sum_{j=1}^s |\hat{w}_{j,\phi}| \|\Gamma_{\hat{r}_{j,\phi}} - \Gamma_{\hat{r}_{j,\psi}}\|_H + \sum_{j=1}^s |\hat{w}_{j,\phi} - \hat{w}_{j,\psi}| \|\Gamma_{\hat{r}_{j,\psi}}\|_H \end{aligned}$$

285 By our assumptions and Lemma 4.7 we have that the first term is proportional to  
 286  $\frac{1}{T}\|\phi - \psi\|_H$ . By inequalities (10), (11) and a reevaluation of our estimates (12)  
 287 and (13) given the assumptions, the second term in the above is also proportional  
 288 to  $\frac{1}{T}\|\phi - \psi\|_H$ . Thus for a suitable  $T$  we have a contraction mapping and by  
 289 Contraction Mapping Theorem our algorithm will converge.  $\square$

290 **Remark 6.** The inequalities above are also dependent on the number of trajectories  
 291 and the path separation distance. Specifically, inspection of the above inequalities  
 292 shows that they are propositional to  $sq_{X,T}^2$ . However, the main point of the theorem  
 293 is that assuming those quantities are fixed and/or bounded then if enough data is  
 294 collected i.e. for a sufficient amount of time the algorithm is guaranteed to converge.  
 295 It should be noted that this is a sufficient condition and other conditions may exist  
 296 in which the algorithm converges.

297 **Remark 7.** By default, occupation kernels are universal if the kernels of the RKHS  
 298 are universal. If  $\gamma(t) = y$  for all  $t \in [0, T]$  we note that  $\Gamma_\gamma(x) = TK(x, y)$ . Hence,  
 299 the set of occupation kernels contains the set of kernels.

300 **5. Numerical Experiments.** In Experiment 1, we generated a flow field  $F(x)$   
 301 using a linear combination of Gaussian kernels and generated a set of random points  
 302 and angles to serve as our anticipated dynamics using unit speed. The flow field is  
 303 given by

$$F(x) = \frac{1}{8} \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix}, \quad (14)$$

where

$$f_1(x) = 5 \exp \left( -2 \left\| x - \begin{bmatrix} .25 \\ .25 \end{bmatrix} \right\|^2 \right) - .2 \exp \left( - \left\| x - \begin{bmatrix} .25 \\ .75 \end{bmatrix} \right\|^2 \right) \\ + 2 \exp \left( - \left\| x - \begin{bmatrix} .75 \\ .75 \end{bmatrix} \right\|^2 \right) - 5 \exp \left( -2 \left\| x - \begin{bmatrix} .75 \\ .25 \end{bmatrix} \right\|^2 \right),$$

and

$$f_2(x) = 3 \exp \left( - \left\| x - \begin{bmatrix} .25 \\ .25 \end{bmatrix} \right\|^2 \right) + \exp \left( - \left\| x - \begin{bmatrix} .25 \\ .75 \end{bmatrix} \right\|^2 \right) \\ - 3 \exp \left( -3 \left\| x - \begin{bmatrix} .75 \\ .75 \end{bmatrix} \right\|^2 \right) + \exp \left( - \left\| x - \begin{bmatrix} .75 \\ .25 \end{bmatrix} \right\|^2 \right).$$

304 The results of Experiment 1 are summarized in Figure 1.

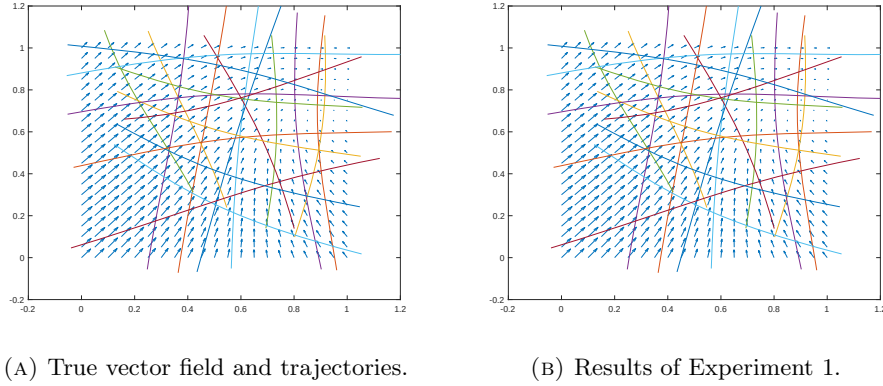


FIGURE 1. Algorithm 1, used for estimation of the vector field in (14). The true trajectories are calculated via RK4 over the time frame  $[0, 1]$ . Using Gaussian RBFs with a kernel width of 1, we performed 10 iterations of Algorithm 1.

305 For Experiment 2, summarized in Figures 2a,2b,2c, and 2d, we used the Glider-  
 306 palooza 2013 data first presented in [5]. The data were for 31 sequential trajectory  
 307 segments and contained the initial positions, the true final positions, dead-reckoned  
 308 final positions, speeds, and dead-reckoning times. For the experiments, the authors  
 309 of [5] used averaged speeds and an average dead-reckoning time of 3.5 hours.

310 Algorithm 1 is designed to work on multiple trajectories necessitating some prior  
 311 calculations. The data was scaled by a factor of  $10^{-4}$ , then the provided initial  
 312 positions and dead-reckoned final positions were used to calculate the initial direc-  
 313 tions along with calculated speeds using their averaged dead-reckoning time of 3.5  
 314 hours. The sequential trajectory data was broken in to 31 separate trajectories.  
 315 Using exponential kernels, we performed 5, 10, and 20 iterations of Algorithm 1  
 316 using a kernel width of  $\mu = 1/170$  for the 5 iteration run and a kernel width of  
 317  $\mu = 1/10,000$  for the 10 and 20 iteration runs. The kernel widths for each of these  
 318 runs were selected to produce well conditioned Gram matrices.

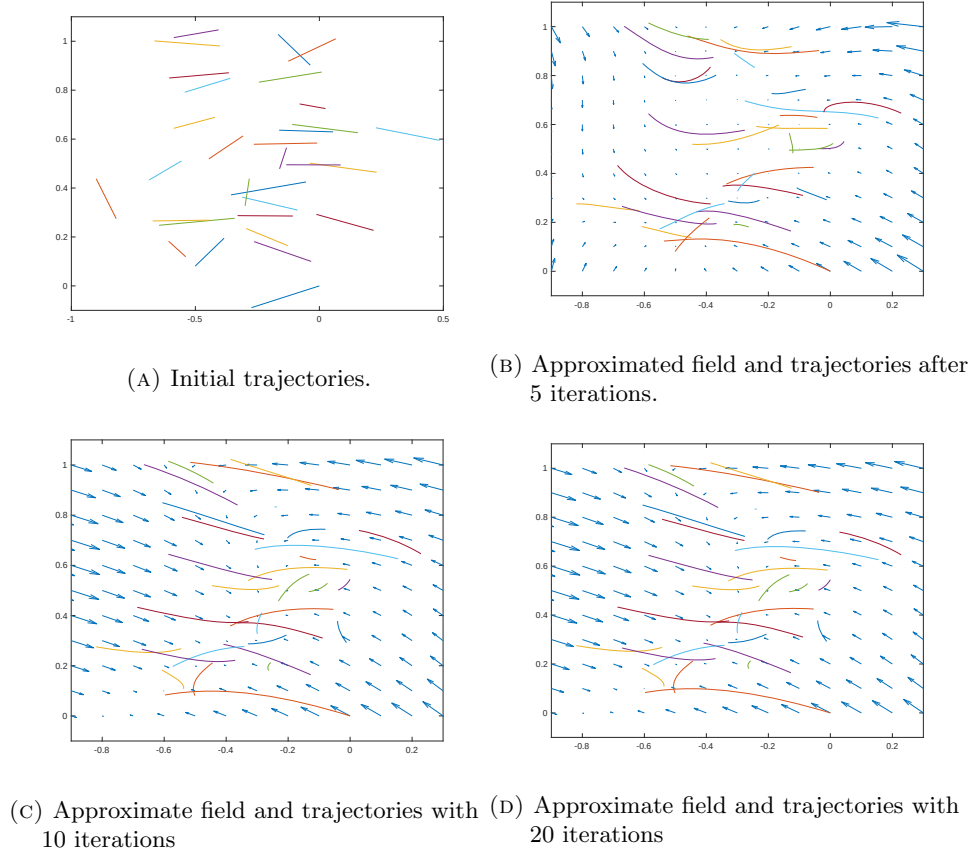
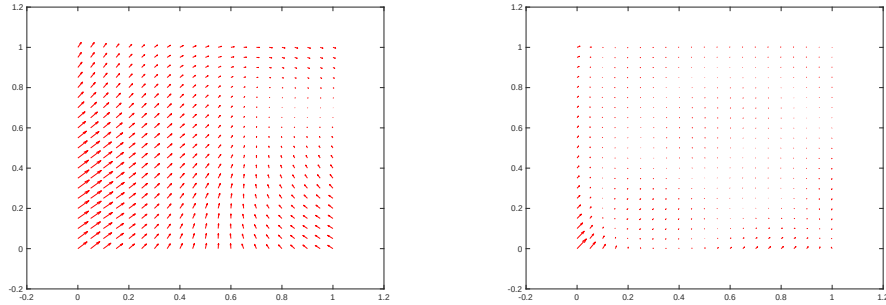


FIGURE 2. Calculated initial trajectories and output of Algorithm 1 for 5, 10, and 20 iterations on Gliderpalooza data.

319 **6. Discussion.** This approach to motion tomography has several advantages over  
 320 that of [5]. In the context of this aim, the flow field is approximated using the  
 321 occupation kernels as basis functions for approximation, whereas [5] requires a piece-  
 322 wise constant description of the flow field or a parameterization with respect to  
 323 Gaussian RBFs. Moreover, [5] renormalizes the integral by multiplying and dividing  
 324 by  $\|\dot{r}(t)\|_2$  to artificially convert the integrals to line integrals. The renormalization  
 325 may ultimately produce divide by zero errors that leads to stronger assumptions on  
 326 the dynamical systems. The occupation kernel method avoids the renormalization  
 327 and does not add further restrictions on the dynamics. Finally, the representation of  
 328 the flow field with respect to the occupation kernel basis allows for the application  
 329 of the approximation powers of RKHSs. It should also be noted that although  
 330 the applications of this technique apply mainly to  $\mathbb{R}^2$  valued functions, there is no  
 331 inherent reason to limit to  $\mathbb{R}^2$ . That is, this technique would apply to  $\mathbb{R}^d$  valued  
 332 functions.

333 Figure 3 compares the output of Algorithm 1 with the method in [5] for estimation  
 334 of the flow field in in Experiment 1. Figures 3a, 3b are the plots of the difference



(A) Difference between the true and approximate field given by the method in [5]      (B) Difference between the true and approximate field given by Algorithm 1.

FIGURE 3. A comparison of Algorithm 1 and the method of [5] for estimation of the vector field in (14).

335 between the true vector field and the approximated vector fields for both methods.

336

337 Due to its structure for spatial representation of the flow-field, the method in [5]  
 338 leads to poor performance for the region where sufficient trajectory information is  
 339 not available. Quantitatively, given a sample of vectors  $\{V(x_1, y_1), \dots, V(x_n, y_n)\}$   
 340 and  $\{W(x_1, y_1), \dots, W(x_n, y_n)\}$  from two vector fields  $V(x, y)$  and  $W(x, y)$ , we can  
 341 define the max error (relative to  $V$ ) as

$$\text{Max Error} = \max_i \{\|V(x_i, y_i) - W(x_i, y_i)\| / \|V(x_i, y_i)\|\} \quad (15)$$

342 and the mean error (relative to  $V$ ) as

$$\text{Mean Error} = \text{Mean}\{\|V(x_i, y_i) - W(x_i, y_i)\| / \|V(x_i, y_i)\|\}. \quad (16)$$

343 Let  $V(x, y)$  represent the true vector field and let  $W(x, y)$  denote the estimated  
 344 vector field. Given the samples shown above, the max and mean errors for both  
 methods are summarized in Table 1.

|            | Method in [5] | Algorithm 1 |
|------------|---------------|-------------|
| Max Error  | 1.1849        | 0.25321     |
| Mean Error | 0.51549       | 0.025642    |

TABLE 1. Algorithm 1, along with the technique in [5] are used to estimate the vector field in (14). The table shows the maximum and the average estimation error, as defined in (15) and (16), respectively, for the two methods.

345

346 Figure 4 shows the the Mean Error as a function of the number of iterations  
 347 for three different vector fields. The first vector field is the one used in the first  
 348 experiment, the second is a linear vector field ( $f_1(x) = x_2, f_2(x) = -0.2x_1$ ) that  
 349 corresponds to a flow that spirals inwards towards the origin, and the third is a  
 350 constant vector field ( $f_1(x) = 0.2, f_2(x) = 0.1$ ). None of the three vector fields are  
 351 in the span of the occupation kernels. The results in Figure 4 demonstrate that the



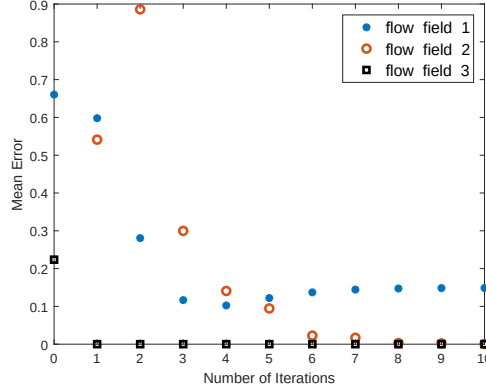


FIGURE 4. To demonstrate universality of the occupation kernel basis, Algorithm 1 is used to estimate three different flow fields. The flow field in (14) (Flow field 1), a linear flow field (Flow field 2), and a constant flow field (Flow field 3). The plot shows the average estimation error, as defined in 16, plotted against the number of iterates of Algorithm 1 for the three flow fields.

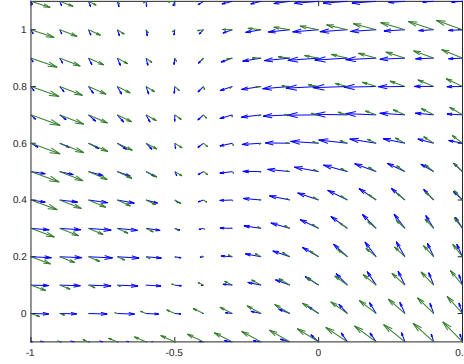


FIGURE 5. Simultaneous plots for estimation of the unknown flow field in the Gliderpalooza experiment using Algorithm 1 (green arrows) and the method of [5] (blue arrows).

352 developed algorithm can estimate a variety of flow fields, not necessarily in the span  
 353 of the occupation kernels. In the absence of ground truth in Experiment 2 we show  
 354 a simultaneous plot of approximated fields from both Algorithm 1 and the method  
 355 in [5] in Figure 5. For the set of samples depicted in Figure 5, we calculated the  
 356 maximum norm difference, the mean norm difference, and the variance of the norm  
 357 differences between the two fields. The results of this comparison are summarized  
 358 in Table 2.

|                      |           |
|----------------------|-----------|
| max norm difference  | 0.14877   |
| mean norm difference | 0.0088628 |
| variance             | 0.0001869 |

TABLE 2. Norm difference statistics between estimates of the Gliderpalooza flow field, approximated using Algorithm 1 and the method in [5].

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